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Transition in South African water governance: Insights from a perspective on power

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ABSTRACT

After apartheid, South Africa has stepped up to the challenge of reforming an inequitable water service delivery system to meet the needs of all citizens. We frame this systematic societal change as a transition in water governance. We argue that when evaluating this pathway of transition, we should not only look at the changes in water legislation and number with improved access, but also analyze the quality of the water service delivery in terms of different payment schemes, participation by local citizens and conflicts around equality of water provision. By analyzing power in transition studies, we explore the power dynamics at play in two regions of Johannesburg, namely Alexandra and Soweto. The paper highlights the need to explicate the politics of water service delivery and suggests opportunities to break the negative patterns in order to achieve equitable and sustainable water service delivery in South Africa.

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1. Introduction

Following apartheid, South Africa was faced with the challenge of redressing the social and environmental imbalances of the past. As a water-stressed country, allocation of water has been tied to development and the changing political landscape of the country. The evolution of water service delivery and the political history of South Africa are inalienably related (Tewari, 2005). In the democratic

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dispensation starting in 1994, particular attention was paid to sectors such as water service delivery where services were previously distributed in a systematically inequitable manner across racial groups, with a specific focus on meeting the needs of whites and excluding Africans (Tewari, 2005).

The new water laws sought to address the social inequities and environmental concerns of the earlier political periods. The National Water Act of 1998 repealed over 100 water acts and related amendments and extinguished all previous public and private rights to water (s. 4 and Schedule 7, RSA, 1998). The significant change brought about by the new legislation was the recognition that water is a scarce and unevenly distributed resource, belonging to all people and no discriminatory law should be established to prevent water access and that sustainability should be the aim in distribution through which all users could derive benefits (RSA, 1997). In 1994, the new government thus had the responsibility to sustainably manage the water resources for the benefit of all people according to the constitutional mandate (s. 3, RSA, 1998).

We conceptualize the overall dynamic patterns of change in water governance in South Africa since 1994 as a transition toward sustainable and equitable water service delivery. Transitions are systematic, complex and long-term societal changes comprising multiple actors at different scales and levels (Geels, 2011; Loorbach et al., 2011). We approach the distributional effects of the water service delivery reform by analyzing the payments for water services and the mechanisms of service delivery provided in two former townships[2] in the City of Johannesburg, namely Alexandra and Soweto. In these two cases, we particularly examine the exercise of power in the processes of social and legal mobilizations with the aim of providing a contribution to the analysis of power in transition processes.

The paper is organized to provide a brief introduction to the methodology of the research in Section 2. This is followed in Section 3 by a discussion of transition heuristic including different conceptualizations and criticisms of the transition framework. The latter part of the section focuses on the role of power analysis in transition heuristic. Section 4 first traces the transition in water service delivery in South Africa and then addresses whether the transition is in lock-in or acceleration phase by examining the power of agents in shaping water service delivery. Section 5 provides reflections from the case studied and discusses opportunities to strengthen the power dimension in transition studies.

2. Methodology

The analytical framework to study power relations in connection with the water service delivery reform in South Africa was drawn from a literature review of transition studies, and academic writing on the transition framework. This analytical framework is tested through a study of water service delivery reform in the city of Johannesburg. This process offers useful material to study a social change as there has been an explicit transition from the pre- to post-apartheid system of water service delivery. The examination on water service delivery within two former townships of Alexandra and Soweto in the City of Johannesburg represent cases of previously disadvantaged areas, characterized by the inequalities that are the focus of reform process. The main data sources for this study consisted of water policy documents for the Republic of South Africa and reports and papers on the payments for water services in the city of Johannesburg. The mechanisms of water service delivery (i.e., the way water is accessed by households either through standpipes, shared taps or taps in houses) were investigated via interviews and field observations.

Since Alexandra, in comparison to Soweto, is less studied, and limited data is available, we carried out more primary data collection in this case. We had five in-depth interviews in three different parts of Alexandra (namely Tsutsumani, East Bank and Old Alex) over two weeks of fieldwork. The respondents were identified through a stratified sampling approach where we first identified the different water service areas in Alexandra. A random sample was selected within these stratified areas based on people's availability and willingness to talk to the researchers. Some of these interviews were conducted through a translator who was fluent in Zulu and English. In addition, we carried out five in-depth interviews with experts in the field including a manager from the city of Johannesburg, a water

[2] In South Africa, the term township usually refers to the (often) underdeveloped urban living areas that, from the late 19th century until the end of Apartheid, were reserved for non-whites (black Africans, Colored and Indians).

engineer from the Department of Water Affairs and Environment, director of the Water Research Commission, manager of the Alexandra Renewal Project and a development facilitator in Alexandra. These individuals were identified by the authors for their expertise and direct knowledge of water service delivery in the area. Given the rich literature available in the case of water service delivery in Soweto, we only conducted a few narrative walks with the residents of Phiri to gain a better understating of the implication of using pre-paid water meters

3. Theoretical framework

3.1. Transition heuristic

The patterns of change in water governance in South Africa are characterized by a complex and long-term process comprising multiple actors at different scales and levels. This process entails changes in technology, culture, policies, politics, power and economics where a wide range of vested interests are involved to promote particular solutions, policy instruments or packages. The complexity and multi-faceted characteristics of this process, so called transition, requires interdisciplinary frameworks that can explore processes of change from different disciplines and societal perspectives (Geels, 2011; Loorbach et al., 2011). In analyzing the change in the South African water governance system, from an old and inequitable to an (envisioned) modern and equitable water access in the city of Johannesburg, we employ transition heuristic (framework). This is useful in three ways.

First, it provides a frame to discuss differences in perception, ambition and understanding of objectives of transition as normative goals, and to orient these aspirations into efforts in a systematic way (Meadowcroft, 2011). Second, transition heuristic is a flexible framework meaning that different theories and concepts can be combined to explore the intricacy of interactions among critical components of a water governance system, i.e. water legislation, policies, and reforms, civil society, policy makers and water organizations (Geels, 2010). By incorporating different social theories on transitions, a comprehensive picture of change processes can be generated and used to assess the water governance practices. Third, transition heuristic compiles a practical toolbox of participatory techniques to choose from and scale up successful experiments to achieve the governance objectives (Rotmans and Loorbach, 2009).

3.1.1. Conceptualization of transition

From a multi-level perspective, transitions can take place when developments at landscape (macro), regime (meso) and niche (micro) levels move in the same direction (Geels, 2002). Landscape acts as a peripheral structure in which regimes and niches interact with each other. Developments at this level are relatively slow and correspond to the broad societal trends (Geels, 2002; van der Brugge et al., 2005). Landscape pressures can be identified at different scales. Climate change, population, international agreements, and MDGs are some of the driving forces from the landscape level on the current water institutions as regime actors in South Africa. The global and national approaches to these issues form dominant discourses that bound the policies and programmes of water institutions in South Africa. The regime may be defined as any form of rules enabling or constraining human activities within communities such as economic rules, legislation or social conventions (Foxon, 2002; van der Brugge et al., 2005). The regime actors may include different entities of the state, such as the Department of Water Affairs (DWA) at national level, City of Johannesburg and Johannesburg Water at local level and so on that are explained later in the 4th and 5th sections of the paper. Niches include organizations, individual persons or innovations instrumental in the take-off of the new regime (Foxon, 2002; van der Brugge et al., 2005). The niche pressures can be described as the experiments and projects formed by a small group of agents that deviate from the regime. These experiments can be scaled up by different means such as technological innovations or through different types of social mobilizations.

Until recently, there has been limited discussion of the role of power, incorporation of theories of power in transition studies or application of techniques that uncover power dynamics in transition frameworks (Lawhon and Murphy, 2012; Meadowcroft, 2011). Although power and politics are key aspects of any change process, most research has focused on socio-technological transitions with

less emphasis on socio-political transitions. Neglecting power in transition analysis brings the risk of ignoring important determinants of social change, and the distribution of costs and benefits within society. Hence, in this paper, we focus on the niche experiments in a more socio-political context than a socio-technological one by exploring the political dimension of transition and the interplay of power between and within different levels.

From a multi-phase perspective, the transition trend is the consequence of shifting from one dynamic state of equilibrium to another. This conceptualization of transition as an S-shaped curve is built on diffusion of innovations theory wherein the process of system transformation or regime-shift unfolds over time (de Haan and Rotmans, 2011). Comprising four main phases, *pre-development phase* is the first stage of transition where the marginal changes happen in the system's background and do not have any impact on the status quo. Second is a *take-off phase* in which the structure of the system starts to slowly change. Third is an *acceleration phase* where the new pattern of the system becomes visible because of accumulation of the changes in the previous stages. Finally, there is a *stabilization phase* where the rate of fluctuation is marginal, and the net effect of any changes is neutral (van der Brugge et al., 2005). We look at the indications of temporal phases not only in terms of progressive changes in water legislation but also their actual and practical implementations in our cases as illustrated in the fourth section of the paper.

Although the multi-phase perspective is often portrayed as a linear, progressive trend, the conceptualization does acknowledge that trends can stall or be reversed. Within the transition heuristic, it is assumed that transitions (toward acceleration) are driven by two mutually reinforcing mechanisms. First is the destabilization of the dominant regime due to landscape pressures, and second is the emergence and up-scaling of niche experiments, e.g., in terms of policy, behavior, technology, etc. (van der Brugge and Rotmans, 2007). In the absence of one of these mechanisms, transition pathways may turn into a lock-in situation. In transition studies, lock-in is considered as one of the transition pathways in which the regime is able to maintain and reproduce its internal dynamics (Geels, 2002; Rotmans and Loorbach, 2009). This can be seen as an exertion of power by the current regime to prevent a change in the status quo. As a result of this process, a set of actions and strategies that could be taken in the future will be limited or constrained by the current regime practices (i.e. actions taken by the regime actors, including institutions). In addition to this "non-ideal" transition pattern, there is a risk of "reverse" transition wherein niche experiments fail to become the new mainstream, and the system will return to its earlier state. This "backlash", is identified as a possible transition pattern and a special case of lock-in (Geels, 2002; Rotmans and Loorbach, 2009). It is suggested that it is at these points that overt conflict occurs and the influence of different actors diverges. Using a multi-phase perspective can offer an opportunity to identify moments of contestation and changing power relations during a transition.

3.1.2. Criticisms

Transition heuristic has been subject to various types of criticisms. Firstly, the framework is criticized for its hierarchical approach of employing landscape, regime and niche levels (Shove and Walker, 2010). The critics have argued for a multi-relational model of analysis emphasizing the notion of horizontal relations between niches, regimes and landscape (Shove and Walker, 2010). They point out that the "nested hierarchy" concept misses out the importance of local practices whose horizontal/flat circulation can be the trigger for transition (versus vertical/hierarchical niche–regime interaction). In response to this, recent work on transition heuristic states that levels (micro–meso–macro) are different degrees of structuration of local practices relating to different scales, and they are not necessarily hierarchical (Geels, 2010, 2011). The important point of this criticism is that one should not only look at the niche networks under the regime level to understand how niches scale up, or how the regime reproduces itself. It is equally important to scrutinize the interaction within a network of heterogeneous niches and how they communicate with each other. We attempt to present the niche interactions with regime actors and other niche actors through a lens of power in the cases presented.

Secondly, the concept of regime and its application to empirical cases is questioned (Berkhout et al., 2004; Genus and Coles, 2008; Markard and Truffer, 2008). Geels (2011) states in response that the regime is an interpretive analytical concept open to investigation from different angles (Berkhout et al., 2004; Genus and Coles, 2008; Markard and Truffer, 2008). In an attempt to specifying the concept, Holtz

et al. (2008) define different characteristics of a regime, such as purposefulness, coherence, stability, autonomy, and heterogeneity of actors. These attributes, in sum, indicate that regimes are autonomous and stable societal systems that serve one or several societal functions bearing on human needs while comprising multiple and heterogeneous actors (Holtz et al., 2008). As societal systems, regimes are also shaped by and shape power relations between actors. The concept of water governance, in its very broad sense, overlaps with this definition of water regime. While we strive to contribute to the development of the concept of water regime through its application in our cases, it is fair to use water governance system as an equivalent not only in terms of technological dimension but also institutional and socio-political aspects as explained in the cases

Third, the transition framework has been criticized for underplaying the role of agency in transitions (Smith et al., 2005). There is a concern that transition heuristic is too structural and therefore leaves little room for the analysis of agency. Geels (2011) responds by suggesting the multi-level perspective (of a transition) is "shot through with agency" (Geels, 2011) and that social changes are only possible through the actions of agents. Furthermore, he acknowledges that certain types of agency are less developed in the transition heuristic, including power struggles.

Finally, the transition framework is also criticized because it incorporates certain theoretical positions, namely evolutionary and structuration theories[3], to analyze the interplay of actors at different levels (Geels, 2010). Some critiques specifically draw attention to oversight of other social theories, particularly insights from conflict theory in capturing the interaction between and within different levels (Avelino and Rotmans, 2009; Lawhon and Murphy, 2012; Smith et al., 2005). The transition heuristic is however a middle range theory that can accommodate different theoretical positions (Geels, 2010). Although it is fair to say that a great deal of research in transition studies has a strong reliance on the theories noted earlier, this does not necessarily exclude the possibility of using other theories to explain the dynamic of transition processes. In responding to this criticism, we aim to contribute to the emerging debates on power in transition studies.

3.2. *A perspective on power dynamics and transitions*

In the first editorial of the Journal of Power, Haugaard and Malesevic (2008) discuss the ubiquity of power and raise a fundamental challenge to the study of power, namely, that it is "all around us, part of the everyday, and hence invisible to the taken-for-granted natural attitude of social practice" (Haugaard and Malesevic, 2008). This has resulted in conceptual and theoretical diversity in the examination of power.

Power can therefore be viewed as an example of Wittgenstein's term 'family resemblance concept' so that when we use the concept in different contexts, its meaning changes sufficiently so that there is no single definition of power which is covered by all usage (Haugaard, 2002). Haugaard (2002) identifies a broad classification of power theory which includes normative political theory of the analytical conceptual variety (Barry, Dahl and Morriss); political theory building of a non-conceptual variety (Aristotle and Arendt); and social theory of modern (Marx, Weber and Bourdieu) and postmodern orientations (Nietzsche and Foucault). This classification offers different language games which theorists employ. The challenge in entering the power debate lies in moving between the different theoretical approaches to power whilst being cognizant of the subtle variation between them. For a summary of the different theoretical works, *see* (Avelino and Rotmans, 2009; Haugaard, 2002; Murphy, 2011). This challenge of terminological subtlety as well as the fact that a great deal of power theory has a high level of abstraction seems to have deterred researchers from engaging with the complexity of analyzing power in transitions (Inglis and Bone, 2006). The research study "Power in Transition" is an attempt to elaborate power dynamics in relation to the transition framework (Avelino and Rotmans, 2009, 2011).

[3] The former assumes the core drivers for transitions are based on entrepreneurs and radical innovations leading to transformation of lock-in regimes (Schumpeter, 1939 in Geels, 2010). The latter, structuration theory, holds the assumptions that actors are knowledgeable agents that actively interpret the rules constraining their actions, and therefore they are capable of re-structure the currently established arrangements through their cognitive capitals, i.e. competences, skills, knowledge, etc. (Giddens, 1984 in Geels, 2010).

Avelino and Rotmans' approach to power can be characterized as lying within the realm of social theory of modern and postmodern orientation. The framework attempts to explicitly conceptualize the role of power in transition studies and connect the study of power to material realities (Avelino and Rotmans, 2009). The framework draws on the work of power theorists, most notably, Luhmann (1995) who views power as a social medium (Luhmann, 1995). Drawing on Parson's (1967) definition, Avelino and Rotmans (2009) define power as "the ability of actors to mobilize resources to achieve a certain goal". A departure from Parson's definition lies in the idea that resources are mobilized by actors rather than by the system and that the goal could be either collective or in one's self-interest (Luhmann, 1995; Parsons, 1967).

Resources are defined broadly by Avelino and Rotmans (2009) and can include persons, assets, materials or capital including human, mental, monetary, artifactual and natural resources. Each type of resource can be the object of power to more or less extent. The resources are themselves considered 'power neutral' but become 'power-ladened' when they are mobilized by actors to reach certain goals (Avelino and Rotmans, 2009).

How resources are mobilized is explained in the framework by looking at a typology of power exercise. Here, different ways in which one can mobilize resources and the different levels these are enacted are used to distinguish five types of power (Avelino and Rotmans, 2009).

- Innovative power: the capacity of actors to create or discover new resources. An example includes the development of new institutions for water management such as the river basin commission or catchment management agency.
- Destructive power: the ability to destroy or annihilate existing resources. An example includes ability of upstream users to restrict water access for downstream users thereby destroying their water supply from rivers.
- Constitutive power: the ability to constitute (establish, institute or enact) a distribution of resources. An example includes transboundary river agreements which create new ways of managing water that extend beyond national institutions.
- Transformative power: the ability to transform the distribution of resources, either by redistributing resources and/or by replacing old resources with new resources. An example includes the constitution of river basin organizations which re-organized the distribution of water resources and institutions around the physical river resource. This changed the way water management is viewed.
- Systemic power: the 'combined' capacity of actors to mobilize resources, for the survival of a societal system. An example includes the system of water rights and permits that may exist in a country to allocate water for different uses. The system only works because of the mobilization of resources by actors within the system to support its functioning, even if the goals of individual actors may not represent a collective goal.

Both regime and niche level actors can exercise power. In this exercise, it is possible to identify two dimensions, namely, the nature of mobilization: constructive versus deconstructive and the level of mobilization: resources versus the distribution of resources (Avelino and Rotmans, 2011). Constructive mobilization brings in resources through innovative power and the distribution of resources is through constitutive power whereas deconstructive mobilization brings in resources through destructive power and distribution of resources is influenced by transformative power. The former is a process of creation and building up new systems whilst the latter is a process of destruction and breaking down of old systems. In the process of applying different forms of power, regime and niche actors are able to influence a transition process. For example, niche actors may apply innovative power to produce alternatives to the status quo whilst regime actors may apply constitutive power to work to maintain the status quo. In mobilizing resources to exercise power, actors must have access to resources; strategies to mobilize resources; skills to apply those methods; and a willingness to do so (Avelino and Rotmans, 2009).

According to Avelino and Rotmans (2011), their most important lesson for transition studies, and 'sustainability governance' more generally, is that antagonistic power dynamics and relations (i.e. overt

critical debates, resistance, disobedience and political conflict) are essential elements for bringing about change and making 'transitions' occur.

3.2.1. Criticisms

Avelino and Rotmans have offered one of the first explicit attempts to conceptualize an analysis of power in transition research. The attempt with this conceptual framework is to move away from the abstract and introduce useable definitions of concepts of power. The intention is to offer a framework that explicitly encapsulates the range of resources and power exercises so that it can be applied across different types of transitions. Their attempt runs the risk of over-simplification, and it can be criticized for its limitations and biases.

Firstly, the labeling of types of power is done in relation to types of actors. Power is thus seen to exist only in relation to how actors mobilize resources. This is a limited view of power and the typology may be restrictive in application. The definition differs from that of Parsons where resources are mobilized by the system (Parsons, 1967). There is thus greater emphasis in the new framework on the role of agency which downplays inherent power of resources. Secondly, there is a lack of recognition of heterogeneity within levels. There is a risk of creating bundled actor categories when in fact there can be a great diversity of actors at different levels. This may result in ignoring potential conflicts within levels as well as the cooperation between some niche and regime actors. A further limitation of the framework is that resources are viewed as power neutral. This may be simplistic if one considers that resources themselves may have a materiality that affects relations (Bakker and Bridge, 2006).

Given the wide-ranging perspectives on power, there are many additional criticisms that can be raised against the framework. Nevertheless, we believe in the merits of this first attempt, and apply the Avelino and Rotmans framework to test its usefulness in the analysis of the transition of water service delivery in South Africa.

Looking at the different strategies for water service provision in the city, and especially within former townships of Alexandra and Soweto, it is possible to identify different resources and tactics mobilized by niche and regime actors to exert power. This influences the nature of the transition in water service delivery in South Africa and influences the equity of the system.

4. Transition in water service provision

4.1. A historical overview

Southern Africa is overall a water scarce region and projected to become more so over time (Hallowes et al., 2008). During the period of apartheid, water service delivery as well as other services were managed through a system of separate governance (Swatuk, 2010), which led to very unequal levels of service in 1994. The new ANC-led Government faced significant challenges for water services, with approximately 15 million people without access to safe water and 20 million without access to adequate sanitation (DWAF, 2006). In this section, the history of reconstructing the water services sector in South Africa, and thereby in the city of Johannesburg, is traced using a transition framework.

4.1.1. Pre-development phase

The first stage in the transition is the pre-development phase where changing socio-environmental conditions impinge on the regime structure. It is not possible to observe any important change of the regime but its resilience decreases. The political struggle from the 1950s and the shift to a democratic state in 1994 in South Africa, was one of the pivotal elements that imposed stress on the centralized and authoritarian water management system. The management of water in South Africa was in keeping with the apartheid policies of the time and there was thus differentiated access across the city of Johannesburg based on racially created spaces for White, Black, Indian and Colored residential areas (Hemson, 2002). Throughout the struggle period against apartheid, civil liberty struggles were coupled with the demand for legitimate housing areas for Black South Africans, especially in urban centers in South Africa (Desai, 2002). Townships such as Alexandra faced the risk of demolition and the removal

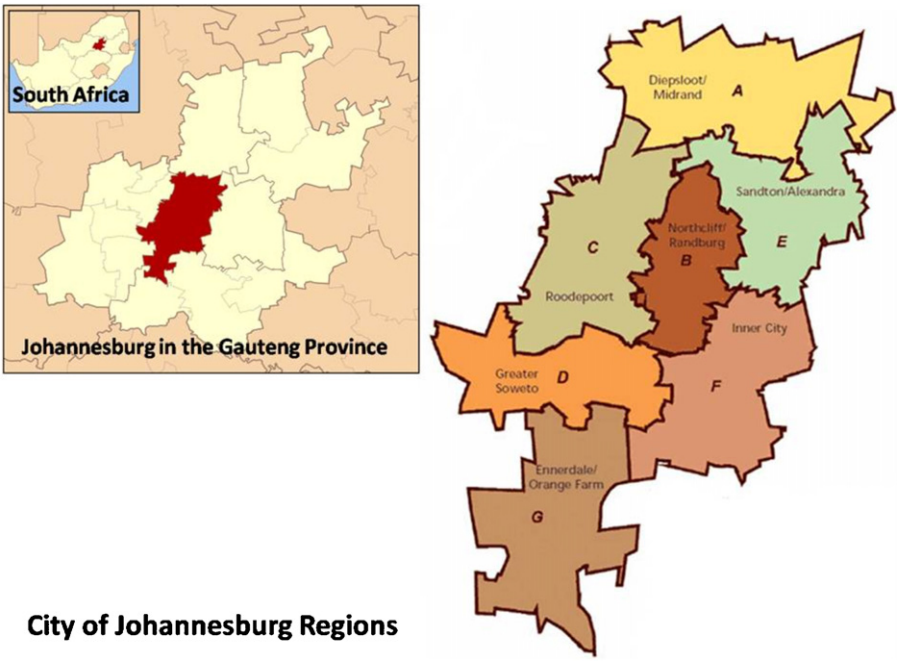

Fig. 1. The Regions of city of Johannesburg including Alexandra, E Region, and Soweto, D Region (City of Johannesburg, 2010b).

of residents to homeland areas or designated townships. In turn, townships formed centers of political activity for the liberation movement.

Post-apartheid, the newly formed Government of National Unity faced the challenge of addressing the inequality in the provision of basic services. Guiding the process was a set of policies that laid out a vision and mission for South Africa's future. The key documents included the Reconstruction and Development Programme (RDP), and the Constitution of the Republic of South Africa (RSA, 1994, 1996). These policies established the sustainability vision (including equitable access to water) in the transition from the apartheid to post-apartheid state. The Constitution states that, every South African has the right to basic access to safe drinking water (RSA, 1996).

4.1.2. Take-off phase

The second stage (take-off phase), in which the structure of the system starts to change, included the creation of new institutions and a new policy environment for water service delivery. The action that led the take-off phase was the introduction of the new National Water Act in 1998 and the policy of free basic water in 2001 (DWAF, 2002). The Constitution provided the basis for these policies. Through the co-operative government model laid out in the Constitution, the duty of water service provision was delegated down to district and local municipalities. This shift required that the City of Johannesburg, as a representative of the state, became responsible for ensuring that all its residents had access to safe drinking water within 200 m from their home (DWAF, 2002). The administration introduced new ways of mapping the city, different from the racial divides of the past. The city has been divided into seven service delivery regions (Fig. 1) including Region E of Sandton/Alexandra and Region D of Greater Soweto which form the focus of this study.

The City of Johannesburg, as the municipal authority is responsible for water service delivery has established Johannesburg Water as its utility responsible for water and sanitation services (City of Johannesburg, 2010b). Johannesburg Water was established in January 2001 as an independent company, the city of Johannesburg being the sole shareholder.

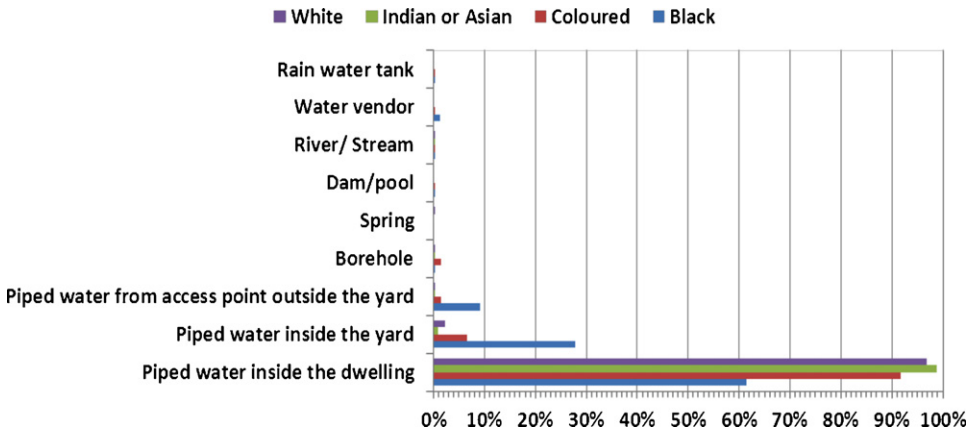

Fig. 2. Access to water by population group of head of household in Johannesburg (Statsa, 2010).

In keeping with global Integrated Water Resources Management principles of managing water as an economic good (Dublin Principles), Johannesburg Water operates under a neoliberal corporate model and provides services along business principles, with the aim of ensuring customer satisfaction and cost recovery (Johannesburg Water, 2011). Annual turnover exceeds ZAR1.6 billion (City of Johannesburg, 2009). The full cost-recovery mechanisms are also endorsed by municipalities as they have been under tight fiscal pressure imposed by national government to be self-sufficient. This pressure is the result of withdrawing central financial support based on landscape level actors such as the World Bank and IMF's advice on decreasing grants and subsidies to local governments (Dugard, 2010). The impact of following the advice has been directly on municipalities' basic services including water and electricity (Dugard, 2010).

As mentioned earlier, water service delivery and infrastructure for different population groups varies across the city of Johannesburg (Fig. 2). Although Johannesburg Water operates as an independent company, the plan for payments for water services is varied according to the needs of residents (City of Johannesburg, 2010a; Gauteng Provincial Government, 2008). Most households in Johannesburg are charged for water on a progressive scale with the cost per kiloliter of water increasing with increasing volumes consumed per month. In all cases, the first 0–6 kL per connection per month are free (City of Johannesburg, 2011). Other households across the city qualify for Expanded Social Package benefits of water and depending on the indigent category they fall into, are able to get an additional allocation of free water per person per day (City of Johannesburg, 2008a). In certain areas, previously deemed consumption areas under the Gcin'amanzi Project, domestic charges are based on a slightly lower sliding scale but water is pre-paid using a metered system (Johannesburg Water, 2006).

Although many new institutions have been introduced, the process of reform there is still a significant challenge to redress apartheid era access which perpetuates according to race as reflected in Fig. 2.

4.1.3. Acceleration or lock-in phase?

The third phase in a transition process is the acceleration phase where the new pattern of structural change becomes visible due to accumulation of socio-cultural, economic and institutional changes in the previous phases. This includes the transformation of power from the traditional regime toward the new regime. At present, it is difficult to say whether acceleration has been achieved or a lock-in situation has occurred. If we consider progressive water legislation as the only indication of the new regime pattern, one could argue that South Africa might be at acceleration stage toward an equitable water service delivery. Nevertheless, the effectiveness of this human rights approach on paper to result in an equitable water provision has been questioned. For example, Bakker (2007), points out the barriers to implementing a "right to water", such as lack of clear responsibility and capacity for

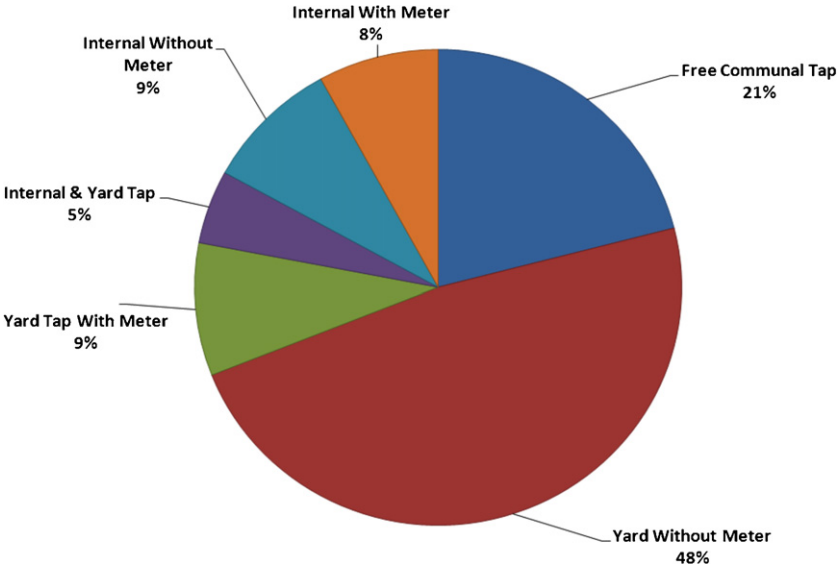

Fig. 3. Water infrastructure in Alexandra (Alexandra Benchmark Survey, 2006).

implementation as well as potential abuse of the concept by governments to over-allocate water to privileged groups at the expenses of other people and also the environment (Bakker, 2007). It is also argued that human right approaches have had little practical impact on tackling the inequality of water distribution in South Africa (Bakker, 2007; Bond and Dugard, 2008). In this sense, one can argue that the regime is in a lock-in situation if institutional changes in water governance have not changed water distribution effectively in term of equality of water access.

As might be seen from the account above, the distinction between different pathways of transition is linked with a set of indicators used to assess the process. We assess the direction of transition not only based on progressive water legislation, but also testing whether there is equitable water service delivery across the city of Johannesburg. Focusing on two areas in the city, namely Alexandra and Phiri in Soweto, we discuss whether the acceleration phase is reached or the transition pathway is in a lock-in situation given the power relations at play between actors at different levels

4.2. Transition dynamics and power in the city of Johannesburg

4.2.1. Water services in Alexandra

Alexandra is characterized by high population density, fast population growth, a young population, elevated levels of unemployment, relatively low levels of education, and low incomes (De Wet et al., 2001). The physical area of Alexandra is divided into 10 unofficial areas for purposes of development initiatives. These areas represent different forms of housing in Alexandra and include formal houses, yards with numerous houses, apartment blocks and informal shacks (De Wet et al., 2001).

A representative of the Alexandra Renewal Project said that all of Alexandra has access to water and the entire area has water infrastructure connections (Fenn, 2010). However there are major differences in water infrastructure amongst different areas. During site visits, we found that households in flats, East Bank and River Park have piped internal water. Those in Setswetla, hostels and Transit Camp share communal taps. In Tsutsumani, households have internal and yard taps.

On paper it seems as if the requirements of water service delivery are being met however the disparity between service levels is significant (Fig. 3). One-fifth of people still rely on free communal taps which require walking from houses to collect water in buckets. A significant improvement was made post-1994 in providing water services by including a tap and toilet on each stand in Alexandra.

However, the Alexandra Benchmark Survey in 2006 as well as our own numeration during fieldwork in 2010 showed that on average there are 19 households per stand in Old Alex. This translates to approximately 133 people sharing the same tap and toilet as compared to 7 people per house in the East Bank or less than 5 in houses in Sandton (where additionally, each house is likely to have more than two taps and toilets).

In Alexandra, we found the full range of payment schemes used by Johannesburg Water with the exception of the pre-paid meters. Some interview respondents paid according to the progressive scale, others benefited from the free water allocation under the Expanded Social Package whilst still others used communal taps. People in the informal settlement of Setswetla were also observed to be using the Jukskei River for washing needs. This is in keeping with the Alexandra Benchmarking Survey which found that household expenditure on water and electricity varied significantly across the different housing areas with only 4% of people living in Marlboro warehouses spending money on water and electricity (average expenditure of ZAR 6) while over 99% of households in East Bank included expenditure on water and electricity (average expenditure of ZAR 251) despite over 90% of households in Alexandra being connected (ARP, 2005). Interestingly, only 14% of households had ever had their water or electricity disconnected for non-payment.

Overall, our findings suggest that water service delivery has improved in Alexandra but the improvement is unevenly distributed across the former township and more so when one considers Region E as a whole. Payment for services is not consistent in Alexandra.

Although there was variability in the distribution of water service infrastructure within Alexandra, it was interesting to note that all interview respondents expressed satisfaction with the level and mechanisms of water service delivery (note, the interviews did not include residents from Setswetla but did cover the other main housing forms in Alexandra). Considering how varied access was, we as outsiders anticipated that people would express a wish to have the higher level of service delivery represented by the tapped water in formal housing structures found in East Bank. As this was not the case, one can ask why this situation is perceived as acceptable. The target of access within 200 m of a household makes perhaps sense in South Africa because of the limited water service during apartheid. However, the innovation of 200 m access enabled the further entrenchment of inequality of access and potentially obstructed the replacement of the older disparities by a new system of equal access. Many residents of Alexandra may feel that some tapped water is better than nothing but the level of service is not comparable to former white, colored and Indian areas. This would be an example of innovative power that 'enables' and 'enforces' constitutive power, meanwhile preventing transformative power (Avelino and Rotmans, 2011). The City of Johannesburg and Johannesburg Water have unintentionally reproduced traditional paradigms and structures that are used to reinforce the inequality in the water services system in South Africa. This has counter-productive results in terms of meeting the sustainability aspirations of creating intra-generational equity.

The subject of cost-recovery and payment for water services has been much debated in connection with poor communities across the world (Desai, 2002). We suggest that payments for water services is an important element in the transition in the South African water sector as it determines water access and raises questions of equity. We found that in Alexandra, many households did not pay for water, with some such as those people living in warehouses accessing water illegally according to one of our interview respondents. In an area with very high unemployment rates, inexpensive or free access to water is a significant benefit to households. However, this situation represents a great dependency of citizens on the state for basic needs. Most of the decision making authority to determine financial support and allocate water dispensations is held by the regime actors, City of Johannesburg and Johannesburg Water. According to the model of co-operative government in South Africa the local tier of government can decide how best to allocate resources internally (RSA, 1996). The establishment of the Alexandra Renewal Project also introduces a top-down influence and an additional regime actor into the agenda setting and decision making processes surrounding water service delivery in Alexandra. Both the identification of Alexandra for the renewal project and the decision to subsidize water in the area were made by regime actors and not through any direct intervention from the local residents. One of the interview respondents suggested that she was not aware why residents paid different fees for water. At present, the government seems prepared to subsidize the cost of water to Alexandra but if financial resources are reduced or diverted elsewhere, many households in

Alexandra will be unable to pay for water and therefore lose access. The power of the regime actors to determine resource allocations may prove unsustainable if the financing for water services change in the city. The regime actors in the transition in water service delivery in Alexandra have mobilized natural and economic resources in a constructive manner to constitute a sanctioned discourse and approach to payments for water services and economic cost recovery that is in line with international norms of Integrated Water Resources Management. The state has by far the greater access to resources, strategies to mobilize financial, technical and human resources, skills to apply these resources to water service delivery as compared to citizens and the willingness to do thus creating the conditions for the exercise of power (Avelino and Rotmans, 2011).

By framing the transition in terms of power, it uncovers an imbalance in the relationship between the state and local citizens (regime-niche interactions) in relation to the development processes of Alexandra. Under the banner of the Alexandra Renewal Project, the re-development of the area has been placed into the hands of outside experts, albeit with some engagement with local residents. Citizens' power and ability to mobilize economic and human resources has been reduced whilst the state's systemic power in managing development in Alexandra has expanded. "For destruction to be an act of power, it must be visible" (Avelino and Rotmans, 2009), but this does not necessarily involve violence or physical force. The destruction of resources, in this case, can be exemplified by destroying mental resources (e.g. an idea, a belief or knowledge) that may not be visible from an insider perspective. Residents seem to accept the disparate service levels across Alexandra and did not express a desire for more standardized service delivery. This is in line with the exercise of ideological power which prevents people, to whatever degree, from having grievances by shaping their perceptions, cognitions and preferences in such a way that they accept their role in the existing order of things, either because they can see or imagine no alternative to it, or because they see it as natural and unchangeable (Lukes, 2005). During the apartheid struggle, residents of Alexandra were actively engaged in the development of their community including providing services not provided by the state. In the current situation, regime actors in the form of the Alexandra Renewal Project and City of Johannesburg drive community development and local residents are less cohesive as a community. As outsiders we find that power exercised by regime actors has been destructive in a sense that has ignored the role of niche actors and their capacities have ultimately become neglected. Power has been exercised in creating a sanctioned discourse but also through organizing participation of local citizens through regime-led participatory initiatives. External actors have been brought into the area to lead the development process and decisions regarding resources are made at national and local government levels by regime actors. This finding is in line with the research pointing out that development agendas in disadvantage areas, as in Alexandra, pay little attention to local experience and knowledge (Mayekiso and Bond, 1996). The nature of mobilization of niche actors is thus limited and becomes deconstructive or disempowering.

4.2.2. Water services in Soweto

Soweto, the second case is also a township that was once established on the borders of the city of Johannesburg and has now been incorporated into the City of Johannesburg. Situated in Region D, Soweto has 15% of the city's informal settlements with an approximate total of 12 809 shacks and 1 300 000 people (City of Johannesburg, 2009). Johannesburg Water is responsible for the provision of water and sanitation to Soweto, ranked as one of the highest water consumption areas in Greater Johannesburg (City of Johannesburg, 2009). The situation in Soweto and particularly within the neighborhood of Phiri is used here as a comparison to Alexandra. The management of water service delivery in Soweto is contrasting sharply to Alexandra especially in terms of payment for water services and social mobilization of local citizens.

Although the situations vary greatly between the two areas, a comparison is appropriate given that both lie within the city of Johannesburg and share similar socio-economic characteristics. In theory, they should therefore be managed according to the same strategy and policies of water service delivery by the City of Johannesburg and Johannesburg Water.

In Soweto, similar to Alexandra, water services have historically been of a low standard and efforts were made from the mid-1990s to upgrade and improve water infrastructure. Problems with economic cost recovery rose for Johannesburg Water due to the inability of residents to pay for their water use. Cost recovery was exacerbated by water losses due to old infrastructure in the area. The over-riding

market logic to ensure cost recovery led Johannesburg Water to introduce a different approach to managing water service delivery. The Operation Gcin' Amanzi pilot project was introduced in mid-2001, to physically restrict water consumption in Phiri, one of the poorest suburbs in Soweto (City of Johannesburg, 2008b; Dugard, 2010). Based on the project objectives, a household could purchase additional water credit by means of pre-paid meters if the water consumption exceeded the obligatory free basic water allocation (6000 L of water per household per month or 25 L per person per day of free water) (Johannesburg Water, 2006).

The Phiri water right case in Soweto can provide us with insights on the power relations between the local citizens and the government in the city of Johannesburg. Johannesburg Water successfully promoted the pre-paid meters in all the communities in Soweto through a participatory process that was perceived as flawed (Barnes, 2009). Council employees were the main participants in stakeholder meetings and a majority of households were not consulted at all about the pre-paid meters (Barnes, 2009). The residents in Soweto were informed that the only way of receiving their free basic water allowance and having their debt written off, is through pre-paid meters while the normal credit meters (promoted in rest of Johannesburg) or the diverse subsidized options available in Alexandra weren't offered as viable alternatives (Ruiters, 2007). Following this process, pre-paid meters were installed across Phiri.

As with Alexandra, Soweto residents were active and vocal during the apartheid era as well as during the formation of the new government in demanding their rights to services and infrastructure. Where the inability to pay for water in Alexandra resulted in various packages which allow people to access water for free, in Soweto, the inability to pay became a threat to accessing water at all (Dugard, 2010). Residents that lagged behind with their payments had a weak bargaining position to resist the installation of the meters (Desai and Pithouse, 2004). The introduction of pre-paid water meters could be viewed as a form of destructive power that made the earlier/traditional system of water governance disappear in Phiri. Citizens living in Phiri were no longer treated the same as residents in other former township areas who had a similar system in terms of water provision (unlimited access) and water charges (pay post-use). Instead Phiri residents now had controlled access and a pre-paid system. In the process, Johannesburg Water also applied innovative power by establishing a new system. The new system included the pre-paid meters and can be seen as disempowering to the residents. The residents in this system were perceived as irresponsible and not able to manage their water use themselves (Ruiters, 2007). The neoliberal corporate model of water service delivery in effect worked to marginalize Phiri residents by treating them differently from other residents of the city of Johannesburg (Bond and Dugard, 2008).

Local citizens mobilized to resist the new system of water service delivery. Members of the Phiri community took the City of Johannesburg to court in December 2007 to challenge its installation of pre-paid water meters, and in April 2008, the South African High Court found this practice unconstitutional and wrote that denying the poor access to adequate water "is to deny them the rights to health and to lead a dignified lifestyle" (City of Johannesburg, 2008b; Dugard, 2010). Further, limiting free basic water to 25 L per person was reviewed and changed to 50 L of free water per person per day provided by the option of an ordinary credit-metered water supply (instead of pre-paid) for more use (City of Johannesburg, 2008b; Dugard, 2010).

The City of Johannesburg appealed the decision in the Supreme Court of Appeal, and in October 2009 the court overturned the ruling of the High Court and declared pre-paid meters lawful (Dlamini, 2009). It also ordered that account holders in Phiri, registered as indigent, should receive 42 L of water per day per resident (Dlamini, 2009).

The social mobilization in the case of Phiri initiated local residents and supported by the Anti Privatisation Forum (APF) and other activists, can be viewed as one of the niche experiments and application of innovative power which were later absorbed by the regime.

The residents constructed a legitimate case for the reconsideration of pre-paid meters as a device of restricting human rights. However, through the course of engagement with the City of Johannesburg, this power exercise was weakened through delays, conflict and a legal process. Through this slow process, City of Johannesburg and Johannesburg Water became quite successful in weakening the Phiri social movement through the judiciary's support for Operation Gcin' Amanzi. In August 2003 any obstruction of the project was banned and many activists and 14 residents of Phiri were charged in a

court of law with malicious damage to property. Consequently, the campaign by the Anti Privatisation Forum and its affiliate organizations was weakened since they had to divert their energy to defeat those charges (Dugard, 2010). One can argue that a relation of power existed where the City of Johannesburg could mobilize more resources that the residents of Phiri. The citizen resistance before 2005 only delayed the process of pre-paid meter installations but it was not able to affect the water management strategies at the regime level. From a power relations perspective, this is an example where innovative power at niche level was not sufficient to be constitutive, Or in other words, the destructive power at regime level was able to deconstruct the mobilization at the niche level and create a lock-in situation. The final decision of the South African Constitutional Court, which was in favor of City of Johannesburg in 2009, can be seen as a path dependent pattern toward a lock-in situation wherein the state dictates its *rules of the game.* The case can thus show how bottom-up approaches cannot be effective in moving toward a desired transition pathway in the presence of obstructive power relations and the absence of protected space for niche experiments.

There is however reason for optimism regarding the process of niche empowerment through legal mobilization. The voices of Phiri citizens were heard across the country and have already had an impact on the broader struggle in South Africa (Dugard, 2010). The presence of conflict may bring a potential change to institutions at regime level (Mahoney, 2000). This can be contrasted with Alexandra where there does not appear to be any overt conflict between citizens and the state with regards to water service delivery.

5. Discussion

5.1. Different exercises of power

By comparing the different exercises of power in the cases, we draw some initial conclusions on the nature of the transition in water service delivery in the city. Water service delivery varies greatly across the city and the inequities of the past are still apparent. Poor (and largely black) urban communities living in informal settlements and former township areas have a much lower quality of water supply than wealthier areas. The different water supply options in Alexandra and the different payment schemes across the city, show that the new system maintains inequities. Part of the reasoning given for the differential supply systems includes the ability to pay for services under a liberal model of water governance (Swatuk, 2010). Inability to pay has led to limited water supply options including shared standpipes. This distinction between wealthy and poor is not however uniform and between the two cases of Alexandra and Soweto, we see differences in the municipal water payment schemes. The neoliberal model of supply management has thus created new mechanisms of instituting inequalities within society (Ahlers, 2010; Narsiah, 2002; Shiva, 2002).

In both Alexandra and Soweto, citizens as community actors have been disempowered by the governance approach adopted for water service delivery. In Alexandra, this seems to have made local actors dependent on the authorities with little exercise of power themselves. In Soweto, the resistance to the pre-paid meters did not influence the outcome but community actors mobilized their resources to engage in governance. These examples indicate that even where conflict is not visible, the exercise of power is at play. Ultimately, the transition in water service delivery can be questioned based on the power imbalances between *citizen and state actors* suggesting that local residents of former townships are compelled to accept a standard of water services prescribed by the local government even if they are not satisfied with it.

5.2. Power between and within the levels

The conceptual framework offered by Avelino and Rotmans (2009) presented a useful starting point for applying a power analysis in our case studies. There are several limitations to their conceptual framework and different approaches to power analysis would introduce different nuances to the investigation.

The conceptualization of power in Avelino and Rotmans' framework revolves around the notion of a capacity that can be mainly exercised between regime and niche actors. Although it is useful for

developing an understanding of the power dynamics between the state and the citizens in Alexandra and Soweto, it provides less support to explore the interaction between and among these communities, and hence, it does not provide us with an insight on how these heterogeneous niche actors interact with each other. The framework could therefore benefit from incorporating a multi-relational analysis within levels as discussed in the theoretical framework (Shove and Walker, 2010). This is particularly crucial in studying water service delivery across the city of Johannesburg wherein the power dynamics between the state and the local citizens differ between various areas. Including an analysis of horizontal relations between the communities in Alexandra and Soweto can be valuable for improving the understanding of power dynamics as well as for finding potential entry points for structural change.

5.3. Agents of change in socio-political transition arenas

An important outcome of the power analysis, is bringing to the fore the role that actors play in determining change through the exercise of power. In the examination of water service delivery in Johannesburg, our research shows that actors at the niche and regime levels in transition arenas are pivotal to maintaining or changing a system.

In breaking a lock-in situation, *frontrunners* are essential (Rotmans and Loorbach, 2009). Frontrunners are one of the key change agents of transition processes. They are "niche players" and "change-inclined regime players" with the "capacity to generate emergent structures within the deviant structures" in a virtual network called the *transition arena* (Rotmans and Loorbach, 2009). The arena can be considered as an experimental space in which the actors use social learning processes to acquire knowledge leading to a new perspective on a transition issue and to be better equipped to exercise power against the lock-in regime. As in the case of Soweto, the transition arena does not need to be the exclusive domain of the experts but can be a space where social and legal mobilization can be initiated. Niche empowerment has a crucial role in the transition management process. It is suggested that actively communicating the shared vision and transition pathways into other networks will encourage people to join the innovation network to build joint strategic agendas (Rotmans and Loorbach, 2009). Focusing on actors with particular competencies, creative minds, strategists and visionaries, this arena is meant to empower niche actors to become a threat to the current distribution of resources, or in other words, a threat to constitutive power at the regime level (Avelino and Rotmans, 2009; Rotmans and Loorbach, 2009). In a sense, the outcome of the transition arena can stimulate the formation of new coalitions and networks to exercise innovative power and to shape transformative power.

Frontrunners and their exercise of innovative power can be observed in the case of Phiri. In building a strategic agenda to move social mobilization forward, a number of actors operating as frontrunners were involved in the court case in Soweto. The Socio-Economic Rights Institute of South Africa, Anti Privatisation Forum, Soweto Electricity Crisis Committee and other groups of political activists can be viewed as frontrunners that created a momentum together with Phiri residents to struggle against prepaid meters. It took active communication of these networks through mass meetings, over a couple of years, to stimulate new ways of resource mobilization. This case has relevance for conflicts over service delivery in many other places in South Africa. Whether a transformative power will emerge as a result of these conflicts and will succeed to break-down the existing regime's constitutive power or not, depend on future acts of frontrunners. According to Dugard, the result of proactive litigation by the Anti Privatisation Forum is too soon to be judged, but initial feedback suggests that social mobilization has not been deterred or discouraged (Dugard, 2010).

The experience of exploring the power dynamics in the transition process in the Soweto case, could be applied in relation to Alexandra. An important aspect of the Soweto case is the role of social mobilization in increasing the systemic power of citizens. There are frontrunners in Alexandra such as non-governmental organizations, community based-organizations, e.g., Wynberg Concerned Residents as niche actors and Alexandra Renewal Project as one of the main regime actors that potentially could initiate social mobilization in relation to the water service delivery transition. In the history of Alexandra, such frontrunner organizations have effected change during the apartheid struggle. These organizations played a central role for safeguarding the continued existence of the township during a

period of destruction of many townships and forced removals under the National Party apartheid government. A frontrunner organization at the time was the Alexandra Liaison Committee headed by Dr. Sam Buti who led a protest against forced removals and was able to "save Alexandra from extinction" (Mafenya, 2002). Such locally based community organizations have been powerful frontrunners in the past and could be so in the future. There is thus a role for social mobilization to be a potential cause of change in water governance in South Africa (Kolb, 2007). This is in line with work by Pithouse (2008) which shows that shack dwellers are constituting a major challenge to technocratic conceptions of democracy (Pithouse, 2008).

In safeguarding the long-term orientation and goals of a transition process, building up continuous pressure on lock-in regimes is *a* central theme and challenge in transition and power studies. Understanding social change processes as transitions can thus allow one to find trigger points for change in stalled processes. Litigation processes may give frontrunners the protected space that they need to create new coalitions and to experience human capacity to mobilize resources to reach certain goals. The continuous process of engaging niche and change-inclined regime actors can create a portfolio of transition experiments that move the struggle against unsustainable water access forward. In doing so, exploring power relations can prevent the constitutive power of the current regime from derailing the struggle.

6. Concluding remarks

Given the process of water service delivery in Alexandra and Soweto in the city of Johannesburg, we argue that the transition pathway is in the take-off phase but not near acceleration phase. Considering the ongoing struggle between niche–regime actors in Soweto and lack of any dynamic or struggle in the case of Alexandra, one could argue that the transition pathway may be closer to lock-in situation where regime actors have been able to *absorb* certain socio-political niche experiments. In our assessment we focused more on socio-political arenas than socio-technological ones. We argue that in evaluating the pathway of transition in water governance we should not only look at the water legislation and the numbers in relation to accessibility to potable water, we must also focus on the conflicts and contestations concerning the justice of the system.

Hence, the power dynamics associated with the post-apartheid development process were focused on here with particular attention to payment for water services as one of the main mechanisms for changing distribution of water across the city of Johannesburg. Understanding the power dynamics at play in water governance is crucial for interventions for strengthening the objectives of equitable and sustainable water access in the city. In relation to the role of citizens and social/legal movements, we point toward how niche-experiments can potentially be scaled up to challenge the regime practices. Delineating the concept of frontrunners and identifying key niche-actors is an attempt to communicate the future transition pathway of water provision into the social networks and build strategic agendas by encouraging people to join these networks.

Incorporation of an analysis on power was not only useful for developing a comprehensive understanding of the ongoing transition, but also beneficial to address one point of critique of the transition heuristic, namely the absence of a discussion of power in analyzing the interplay of actors at different levels. We have offered a contribution to the ongoing debates on framing power by looking at existing work in transition studies. Our research suggests that initial attempts are in the right direction but there is a great deal more room for the political dimension of transitions to be developed to fully capture the dynamics of societal change.

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